Inference-Preserving Error-Bounded Compression for On-Device Machine Learning in Wearable Health Systems

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Abstract—On-device machine learning for wearable health systems must balance inference fidelity against the energy cost of transmitting high-frequency physiological signals, and wireless transmission dominates the power budget of battery-operated body-area networks. We ask how much of the transmitted signal is actually needed for accurate inference, and answer it with an error-bounded lossy compression pipeline (closed-loop Lorenzo prediction, uniform quantization with bin width 2ϵ, and Huffman coding) that is light enough for a microcontroller and provides a formal pointwise guarantee $|\hat{x}_i - x_i| \leq \epsilon$. We evaluate it on the four wrist-worn Empatica E4 channels of the WESAD dataset (BVP 64 Hz, EDA 4 Hz, TEMP 4 Hz, 3-axis ACC 32 Hz; 15 subjects), using a feed-forward neural network for binary stress detection under leave-one-subject-out cross-validation, with scalar quantization, downsampling, ZFP and lossless coding as baselines. Every reported number is produced by the released code. Up to $\epsilon = 0.05\sigma$ the pipeline reaches rate-weighted compression ratios of 8.3× to 17.3× while LOSO accuracy stays within one point of the uncompressed 91.6% (no difference significant at $p < 0.05$); at $\epsilon \geq 0.1\sigma$ ratios of 20.5× to 23.2× cost about three points. Downsampling matches this accuracy only up to 8× and degrades sharply without retraining, and ZFP reaches at most 5.5× at the same bound. The measured bound-to-accuracy curve gives system designers a single tunable, interpretable knob linking radio energy to inference quality.

Index Terms—On-device ML, wearable sensing, lossy compression, edge inference, stress detection, WESAD, energy efficiency, TinyML

I. Introduction

Wearable health monitoring systems increasingly rely on continuous physiological sensing to support real-time inference tasks such as stress detection, arrhythmia identification and activity recognition [1]. These systems generate sustained multimodal data streams that must be relayed from a body-worn sensor node to an edge device for analysis, and as always-on monitoring becomes standard the energy cost of this link becomes a primary constraint on system viability. In battery-powered sensor nodes, wireless transmission is the dominant energy consumer, often exceeding the cost of sensing and computation by one to two orders of magnitude [2].

This asymmetry motivates a simple question: how much of the transmitted signal is actually necessary for accurate inference? If the reconstruction error introduced by compression stays below the level at which discriminative features degrade, communication can be reduced substantially without a meaningful loss in model performance. Prior work on edge ML either assumes lossless transmission [4] or applies generic compression without measuring the downstream effect, while the scientific-computing community has developed error-bounded compressors such as SZ [6] and ZFP [7] that provide formal ℓ_∞ guarantees but have not been evaluated inside a wearable inference pipeline.

This paper bridges that gap with three contributions. (1) A device-level, SZ-inspired error-bounded compression pipeline for multivariate wearable signals whose predictor runs on reconstructed samples, so the ϵ guarantee holds by construction and is verified numerically on every sample of the dataset. (2) A systematic evaluation of the compression–accuracy trade-off on the WESAD stress-detection benchmark [8] under leave-one-subject-out (LOSO) cross-validation with paired statistics across subjects, against scalar quantization, downsampling, ZFP and lossless baselines. (3) An open, fully reproducible protocol in which the paper's tables are generated directly by the experiment code.

II. Related Work

Compression in sensor networks. Razzaque et al. [3] survey two decades of compression for resource-constrained sensor networks and find that lossy methods consistently outperform lossless ones in ratio while remaining adequate for monitoring applications. Transform-based coding (DCT, wavelets) has been applied to biosignals such as ECG [5], but its computational cost is high for ultra-low-power microcontrollers.

Error-bounded lossy compression. Error-bounded compressors guarantee that $|\hat{x}_i - x_i|$ never exceeds a user-specified ϵ. The SZ family [6] uses a predict–quantize–encode pipeline with the Lorenzo predictor and quantization bins of width 2ϵ; ZFP [7] uses orthogonal block transforms with fixed-rate or fixed-accuracy encoding. Both target HPC simulation data, and Azar et al. [10] showed that a lightweight SZ variant runs on IoT microcontrollers.

Edge ML for wearable health. Li et al. [4] identify communication cost and inference latency as the dominant constraints of deep learning at the IoT edge. WESAD [8] is the most widely used wearable stress benchmark; reported accuracies range from roughly 80% to 92% depending

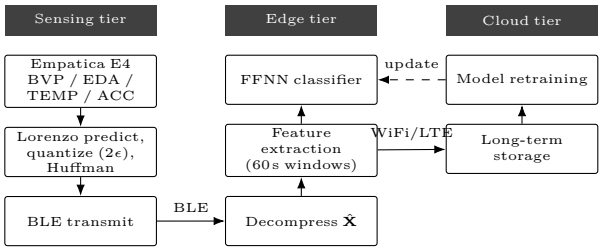


Fig. 1. Three-tier architecture with on-device error-bounded compression.

on modality and protocol [9], but the effect of upstream lossy compression on downstream accuracy has not been quantified.

III. System Model

A. Architecture

Fig. 1 shows the three-tier architecture. The sensing tier (Empatica E4) acquires the signals and runs the compression pipeline on a transmit window of $T_{\text{tx}} = 60\,\text{s}$; the edge tier (smartphone or gateway) decompresses the bitstream, extracts window features and runs the classifier; the cloud tier stores data and retrains the model. The classifier therefore always operates on reconstructed signals, which is the matched scenario evaluated below.

B. Energy model

For a radio whose transmit energy per window is E_{tx} at the uncompressed rate, compression by a ratio ρ (uncompressed bits over transmitted bits) reduces the per-window transmit energy to

$$E_{\text{tx}}^{(c)} = E_{\text{tx}}/\rho, \tag{1}$$

assuming the radio duty cycle scales with payload size. The uncompressed representation is one IEEE float32 per sample, i.e. 168 samples/s × 32 bits = 2.42 MB per hour for the four E4 channels; we also report bits per sample, which is independent of that choice. The compressor itself requires one multiply-add, one division and a table lookup per sample, which is within the budget of an ARM Cortex-M class core [10].

IV. Compression Framework

A. Closed-loop prediction and quantization

Each channel is processed independently. For sample $i \geq 3$ of a window, the 1-D Lorenzo predictor extrapolates linearly from the two previously reconstructed samples,

$$\tilde{x}_i = 2\hat{x}_{i-1} - \hat{x}_{i-2}, \tag{2}$$

the residual $r_i = x_i - \tilde{x}_i$ is quantized to the nearest multiple of $q = 2\epsilon$, $k_i = \lfloor r_i/q + \frac{1}{2} \rfloor$, and the reconstruction is $\hat{x}_i = \tilde{x}_i + k_i q$. Because the decoder forms the identical $\tilde{x}_i$ from the same reconstructed history, $|\hat{x}_i - x_i| = |k_i q - r_i| \leq q/2 = \epsilon$ for every sample. The first two samples of each window are stored as quantized absolute values (32-bit indices), and any residual whose index would exceed $2^{15}-1$ is escaped and stored as a quantized absolute value, so the bound holds unconditionally. An earlier draft of this work predicted from the original samples x_{i-1}, x_{i-2}; the decoder cannot form that prediction, quantization errors then accumulate through the recursion, and we measured peak errors of 3.1×10^8 to 6.8×10^9 times ϵ on WESAD, which is why the closed-loop form of (2) is essential.

Algorithm 1 On-device compression of one transmit window

Require: window $\mathbf{X} \in \mathbb{R}^{M \times N}$, per-channel bounds $\epsilon_c > 0$
Ensure: bitstream $\mathbf{b}$

```
for each channel c = 1 to N do
    q ← 2ϵ_c;   x̂_1, x̂_2 ← q⌊x_{1,2}/q + 1/2⌋ (stored)
    for i = 3 to M do
        x̃_i ← 2x̂_{i−1} − x̂_{i−2};   k_i ← ⌊(x_i − x̃_i)/q + 1/2⌋
        if |k_i| > 2^15 − 1 then escape: x̂_i ← q⌊x_i/q + 1/2⌋
        else x̂_i ← x̃_i + k_i q
    end for
    Huffman-encode {k_i} with a per-window canonical table; append table and codes to b
end for
return b
```

B. Choosing ϵ across heterogeneous channels

The four E4 channels have very different physical scales (BVP in arbitrary units of order 10^2, EDA in μS, TEMP in °C, ACC in 1/64 g). A single absolute bound would be meaningless across them, so we express the bound per channel as a fraction of a fixed scale constant, $\epsilon_c = \epsilon\,\sigma_c$, where σ_c is the global standard deviation of channel c over the corpus (BVP 65.2, EDA 2.34 μS, TEMP 1.5 °C, ACC 44.1). These are label-free constants stored in firmware; the resulting ϵ_c are absolute, known in advance and reported in Table III. We sweep $\epsilon \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$.

C. Entropy coding and bit accounting

Residual indices are Huffman-coded per transmit window and channel. All overheads are counted: a 16-bit length, two 32-bit boundary indices, a canonical code table (16 bits per symbol plus 6 bits per code length) and 32 bits per escaped sample. The compression ratio is the total uncompressed bits divided by the total transmitted bits over all channels, subjects and windows; this rate-weighted ratio is what determines radio energy, and it is dominated by the high-rate BVP and ACC channels.

D. Baselines

Scalar quantization uses the same bins $q = 2\epsilon_c$ without prediction, coding each window with the cheaper of fixed-length packing and Huffman coding (also error-bounded). Downsampling keeps every D-th float32 sample ($D \in \{2, 4, 8\}$, ratio exactly D) with linear interpolation at the receiver; it has no error bound. ZFP [7] runs in

TABLE I
Compression ratio and LOSO stress-detection accuracy of the proposed pipeline (matched scenario; mean ± 95% CI over 15 subjects; p: Wilcoxon vs. uncompressed).

Config.	ϵ/σ	Ratio ρ	bits/samp.	Acc. (%)	ΔAcc.	p	F1 (%)
Uncompressed	—	1.0×	32.00	91.6 ± 4.2	—	—	89.9
Proposed	0.005	8.3×	3.85	91.4 ± 3.8	-0.3	0.25	89.4
Proposed	0.01	10.3×	3.12	90.7 ± 4.3	-1.0	0.06	88.5
Proposed	0.02	13.1×	2.45	91.9 ± 3.8	+0.2	0.88	90.0
Proposed	0.05	17.3×	1.85	91.2 ± 4.0	-0.5	0.75	89.0
Proposed	0.1	20.5×	1.56	88.6 ± 3.5	-3.0	0.04	85.8
Proposed	0.2	23.2×	1.38	88.2 ± 2.9	-3.4	0.06	85.4

fixed-accuracy mode with tolerance ϵ_c on each whole per-subject channel stream, which avoids its per-stream header overhead and favours the baseline. Lossless coding applies zlib (level 9) to the float32 bytes.

V. Experimental Setup

Data. WESAD [8] provides synchronized recordings of 15 subjects (S2–S11, S13–S17) in baseline, stress (Trier Social Stress Test) and amusement conditions. We use only the wrist-worn E4 channels. Windows of 60 s with a 10 s stride are labelled by the majority condition if its purity exceeds 90%, giving 3101 windows of which 926 (29.9%) are stress; the task is binary stress vs. non-stress (baseline + amusement).

Features. From each (reconstructed) window we compute 47 standard features: BVP statistics, heart rate and heart-rate variability from band-pass-filtered BVP peaks (mean HR, SDNN, RMSSD, pNN50) and spectral power ratios; EDA statistics, slope, tonic level and skin-conductance-response count and amplitude; TEMP statistics and slope; and per-axis and magnitude statistics, jerk and spectral peak of ACC.

Classifier and protocol. A feed-forward network with two hidden layers (64, 32 units, ReLU [11], Adam, $L_2 = 10^{-3}$) is trained on standardized features with random oversampling of the minority class. Evaluation is LOSO: for each of the 15 subjects the model is trained on the other 14 and tested on the held-out subject; results are averaged over 5 random seeds. In the matched scenario the model is trained and tested on signals reconstructed at the same ϵ; in the mismatched scenario it is trained on uncompressed signals and tested on reconstructed ones. We report mean accuracy and macro-F1 across subjects with 95% confidence intervals, and the paired difference to the uncompressed baseline across subjects with a Wilcoxon signed-rank test ($n = 15$).

Bound verification. For every error-bounded method, subject, channel and setting, the script records $\max_i |\hat{x}_i - x_i|/\epsilon_c$ and aborts the paper's tables if any value exceeds 1; an independent decoder must also reproduce the encoder's reconstruction bit-exactly.

VI. Results

Bound verification. Across all six settings, all 15 subjects and all channels the worst-case ratio $\max|\hat{x} - x|/\epsilon$ is 0.999999 (Table III): the guarantee held on every one of the 1.5×10^7 reconstructed samples, the quantizer uses its full budget, and the independent decoder reproduced the encoder reconstruction bit-exactly. The incorrect open-loop predictor of the earlier draft exceeded ϵ by eight to nine orders of magnitude on the same data.

TABLE II
Baselines. Top: error-bounded methods at each ϵ (rate-weighted ratio, accuracy change vs. uncompressed in points, and for ZFP the realized worst-case error). Bottom: methods without a tunable bound (ratio, matched ΔAcc., accuracy of an uncompressed-trained model on the reconstructed test data).

ϵ/σ	0.005	0.01	0.02	0.05	0.1	0.2
Proposed: ratio ρ	8.3×	10.3×	13.1×	17.3×	20.5×	23.2×
Proposed: ΔAcc.	-0.3	-1.0	+0.2	-0.5	-3.0	-3.4
Scalar quant.: ratio ρ	5.1×	5.5×	6.8×	9.5×	12.5×	16.5×
Scalar quant.: ΔAcc.	-0.3	-1.0	+0.2	-0.5	-3.0	-3.3
ZFP: ratio ρ	3.0×	3.3×	3.6×	4.3×	4.8×	5.5×
ZFP: realized $\max|\hat{x} - x|/\epsilon$	0.47	0.47	0.57	0.46	0.46	0.51
ZFP: ΔAcc.	-0.2	-0.2	+0.2	+0.7	+0.2	+0.2
Not error-bounded / reference: ratio, ΔAcc. matched, Acc. mismatched						
Downsampling $D = 2$	2.0×, +0.4, 91.4%					
Downsampling $D = 4$	4.0×, +1.0, 89.7%					
Downsampling $D = 8$	8.0×, +0.1, 85.4%					
Lossless zlib	3.2×, -0.0, 91.6%					
Proposed, native resolution	4.5×, +0.0, 91.7%					

TABLE III
Per-channel ratio of the proposed pipeline and worst-case $\max|\hat{x} - x|/\epsilon$ over all subjects and samples.

Channel	σ (unit)	$\epsilon = 0.005\sigma$	$\epsilon = 0.01\sigma$	$\epsilon = 0.02\sigma$	$\epsilon = 0.05\sigma$	$\epsilon = 0.1\sigma$	$\epsilon = 0.2\sigma$
BVP	65.2 a.u.	9.4×	11.9×	14.4×	17.5×	20.0×	22.7×
EDA	2.34 μS	11.6×	13.9×	16.1×	18.3×	19.5×	20.2×
TEMP	1.5 °C	11.8×	15.1×	18.0×	19.2×	20.1×	20.9×
ACC x	44.1 1/64 g	7.5×	9.2×	12.1×	17.1×	21.0×	23.9×
ACC y	44.1 1/64 g	7.5×	9.2×	12.1×	17.1×	21.0×	23.9×
ACC z	44.1 1/64 g	7.5×	9.2×	12.1×	17.1×	21.0×	23.9×
All (rate-weighted)		8.3×	10.3×	13.1×	17.3×	20.5×	23.2×
Max $|\hat{x} - x|/\epsilon$		1.000	1.000	1.000	1.000	1.000	1.000

Compression–accuracy trade-off. Table I shows the uncompressed baseline at 91.6% accuracy (macro-F1 89.9%), within the range reported for wrist-only WESAD classifiers. From $\epsilon = 0.005\sigma$ to 0.05σ the ratio grows from 8.3× to 17.3× while the accuracy change stays between -1.0 and +0.2 points and no paired difference is significant (smallest $p = 0.06$ at 0.01σ); the non-monotonic ordering of these four settings is within the confidence intervals and should be read as noise. Beyond that the trade-off turns: $\epsilon = 0.1\sigma$ gives 20.5× at -3.0 points ($p = 0.04$) and 0.2σ gives 23.2× at -3.4 points ($p = 0.06$). Per channel (Table III) the smooth, oversampled TEMP and EDA streams compress most at tight bounds, while at loose bounds every channel approaches the one-bit-per-sample floor of Huffman coding; BVP and ACC carry 95.2% of all samples and therefore set the rate-weighted ratio.

Comparison with baselines. Table II isolates the contributions. Scalar quantization reconstructs onto the same grid and reaches the same accuracy as the proposed pipeline at every ϵ, so the Lorenzo predictor costs nothing in fidelity and multiplies the ratio by 1.4× to 1.9×. ZFP in fixed-accuracy mode realizes only 0.46 to 0.57 of the permitted error and reaches 3.0× to 5.5× even with whole-stream encoding; its accuracy changes (-0.2 to +0.7

points) reflect that smaller realized error. Downsampling is a strong baseline when the classifier is retrained on downsampled data: $D = 4$ reaches 92.6% (+1.0 points, $p = 0.30$) and $D = 8$ reaches 91.7%, statistically indistinguishable from the proposed pipeline at 8.3× ($\epsilon = 0.005\sigma$, 91.4%, paired difference -0.4 points, $p = 0.89$), and at 4× it is in fact 1.2 points better than the 8.3× setting ($p = 0.03$). The advantage of the error-bounded pipeline is therefore not accuracy at low ratios but that it continues to 17.3× with -0.5 points, roughly twice the ratio of the best downsampling setting, while providing a reconstruction guarantee that downsampling cannot offer. Lossless zlib gives 3.18×, and the proposed codec at the sensors' native resolution (exact for BVP, TEMP and ACC) gives 4.5× at 91.6%.

Mismatched deployment. When the classifier is trained on uncompressed data and deployed on reconstructed signals, accuracy is 91.7% at $\epsilon = 0.05\sigma$ (matched: 91.2%) and 86.5% at 0.1σ, so a model need not be retrained for bounds up to 0.05σ. Downsampling behaves differently: the same uncompressed model drops to 89.7% at $D = 4$ and 85.4% at $D = 8$, because interpolation changes the feature distribution in a way that the bounded quantization error does not.

Energy. By (1), the ratio at $\epsilon = 0.05\sigma$ cuts per-window transmit energy and the hourly payload by 94.2% (1.85 bits per sample instead of 32), and $\epsilon = 0.2\sigma$ by 95.7%; lossless coding of the same float32 stream saves only 68.5%, so most of the gain comes from the controlled loss rather than from redundancy of the representation.

VII. Discussion and Limitations

The guarantee is on reconstruction error, not on classification error; the latter is measured empirically and depends on the features and model. The bound is set relative to corpus-level scale constants, so the absolute bounds must be re-derived for a different sensor; moreover, corpus-level scaling is coarse for EDA, whose per-subject range varies from 0.97 to 13.1 μS, so at $\epsilon = 0.1\sigma$ (0.23 μS) the EDA of low-conductance subjects collapses onto a few levels, which is the most likely cause of the accuracy drop at that setting; per-subject or per-channel bounds are an obvious refinement. Accuracy is evaluated with a single, deliberately simple feature set and network to isolate the effect of compression; stronger models would shift the absolute numbers but not the design of the protocol. With 15 subjects the paired tests have limited power, and the 95% intervals in Table I are wide; differences inside those intervals should not be over-interpreted. Transmit energy is modelled as proportional to payload, ignoring BLE connection overhead, and encoder energy on the E4 was not measured.

VIII. Conclusion

We presented an inference-preserving, error-bounded lossy compression pipeline for wearable health monitoring, with a closed-loop predictor that makes the ϵ guarantee hold by construction, and evaluated it end-to-end on WESAD under LOSO cross-validation against scalar quantization, downsampling, ZFP and lossless baselines. The results show that wrist physiological signals contain substantial redundancy relative to the needs of a stress classifier: ratios of 8.3× to 17.3× leave LOSO accuracy within one point of the uncompressed 91.6%, ratios above 20× cost about three points, and plain downsampling is competitive only up to 8× and only with retraining. The error bound is a single interpretable control parameter linking radio energy to inference fidelity, and all numbers in this paper are generated by the released code.

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